Working Paper: Implementing Generative AI in the Historical Studies

工作论文：在历史研究中实施生成式人工智能

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Abstract

摘要

This working paper provides a detailed overview on implementation strategies, critical evaluation, and best practices to use generative AI for research and analysis in historical studies. It distinguishes between two key applications of AI in historical research: as a supplementary tool and as a methodological component. The paper further emphasizes a responsible, purposeful, and demand-oriented approach with generative AI, which is guided by the principle of minimal computing. It outlines various application scenarios and provides clear frameworks for evaluating when and how to implement AI technology, addressing both individual and institutional responsibilities in order to maintain scientific integrity and adherence to data security and privacy. The aim is to maximize research impact and to maintain space for creative exploration and breakthrough innovation, while balancing performance and reliability needs with ethical considerations and sustainability.

中文译文：本工作论文详细概述了在历史研究中使用生成式人工智能进行研究与分析时的实施策略、批判性评估方法与最佳实践。文章区分了人工智能在历史研究中的两类核心应用：一是作为辅助性工具，二是作为研究方法的组成部分。文章进一步强调，应当以负责任、有明确目的并以实际需求为导向的方式使用生成式人工智能，同时遵循“最小计算”（minimal computing）原则。作者列举了多种应用场景，并提出了清晰的评估框架，用以判断何时以及如何引入人工智能技术；同时从个人与机构两个层面讨论责任，以维护科学研究诚信，并遵守数据安全与隐私要求。其目标是在兼顾伦理与可持续性的前提下，平衡性能与可靠性需求，尽可能提升研究影响，同时保留创造性探索和突破性创新的空间。

This working paper builds upon the reader’s basic understanding of how Large Language Models (LLMs) work in general, and focuses on providing practical guidance for researchers, who plan to integrate AI technology into their historical research but face uncertainties about implementation, critical evaluation, and best practices. Key aspects covered in this working paper include:

中文译文：本工作论文假定读者已经对大型语言模型（LLMs）的一般工作方式具有基本了解，并重点为那些计划将人工智能技术整合进历史研究、但对具体实施方式、批判性评估以及最佳实践仍存在疑问的研究者提供实用指导。本文涵盖的核心内容包括：

- Assessment of the usage of generative AI as tools versus methodological applications (with application scenarios).
  - 评估生成式人工智能作为研究工具与作为方法论应用时的不同使用方式，并提供相应应用场景。
- Risk assessment frameworks covering legal, ethical, and environmental aspects, focusing on scientific integrity, input evaluation, and reflection of outputs.
  - 建立涵盖法律、伦理和环境层面的风险评估框架，重点关注科学研究诚信、输入内容评估以及对输出结果的反思。
- Recommendations for a responsible use of AI technology.
  - 提出负责任使用人工智能技术的建议。
- Recommendations for different implementation levels and types of models.
  - 针对不同实施层级与不同模型类型提出建议。

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中文译文：致谢：我们感谢莱布尼茨欧洲历史研究所（IEG）“数字性”（Digitality）工作组对本文的讨论，也感谢 Fabian Cremer 与 Thorsten Wübbena（按字母顺序排列）在多轮反馈中提出的意见。

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AI as Tool or as a Method?

人工智能：作为工具还是作为方法?

This working paper distinguishes between two implementation levels of generative AI in academic work: its use as a research support tool and its integration as a methodological component. While these classifications naturally overlap and are not always clear-cut, understanding their primary distinctions helps clarify their application and impact in research processes.

中文译文：本工作论文将学术研究中生成式人工智能的实施分为两个层级：一是作为研究支持工具使用，二是将其整合为方法论组成部分。虽然这两类用法自然存在重叠，边界也并非总是清晰，但理解它们的主要差异，有助于明确人工智能在研究流程中的具体应用方式及其影响。

When we refer to AI as a tool, we are describing technology that supports and enhances existing research processes. This includes applications like text translation, content summary, or data organization—tasks that support rather than fundamentally alter research methodology.

中文译文：当我们把人工智能称为一种“工具”时，指的是它用于支持和增强既有研究流程。例如文本翻译、内容摘要或数据整理等应用；这些任务主要发挥辅助作用，而不会从根本上改变研究方法。

In contrast, AI as a method refers to the direct integration of AI technology into the analytical framework. Here, AI becomes part of the methodological approach itself, particularly in source analysis and interpretation. This goes beyond supporting the research processes and becomes an integral part of how we understand and interpret our research materials. To increase clarity, we highlighted recommendations for using AI as a method with violet text throughout this paper.

中文译文：相比之下，“人工智能作为方法”指的是将人工智能技术直接整合进分析框架。在这种情况下，人工智能本身成为方法论路径的一部分，尤其参与史料分析与解释。它不再只是辅助研究流程，而是直接参与我们理解和阐释研究材料的方式。为便于区分，原文中与“人工智能作为方法”有关的建议统一以紫色文字标示。

Fluidity of AI as a Tool and as a Method

人工智能作为工具与方法之间的流动性

It is important to note that the distinction between tool-based and methodological AI implementation is not always clear-cut, and some applications may fall into overlapping categories. This fluidity becomes clear on the example of Optical Character Recognition (OCR):

中文译文：需要注意的是，以工具为基础的人工智能应用与方法论层面的人工智能应用之间，并不存在始终清晰的界线；某些应用可能同时落入两类范畴。以光学字符识别（OCR）为例，可以清楚看到这种流动性：

When used as a tool, OCR might be employed to convert documents or sources into machine-searchable text to facilitate further review. When used as part of a methodology, OCR becomes the crucial first step in a systematic corpus analysis, where the accurate digitization of archival material is essential for subsequent computational text analysis, or natural language processing. The quality and characteristics of the OCR process directly influence the research results and must be considered part of the methodological framework.

中文译文：当 OCR 被作为工具使用时，它可以将文献或史料转化为机器可检索文本，以便后续查阅；而当 OCR 被纳入研究方法时，它就会成为系统性语料库分析中的关键第一步。在这种情况下，档案材料能否被准确数字化，会直接影响后续的计算文本分析或自然语言处理。因此， OCR 过程本身的质量与特征会直接影响研究结果，必须被视为方法论框架的一部分。

Research-based Application Scenarios for Generative AI

面向研究的生成式人工智能应用场景

General application scenarios in the historical studies entail the following non-exhaustive and non-ordered lists of tasks and implementation examples:

中文译文：在历史研究中，生成式人工智能的一般应用场景包括以下任务与实施示例。该列表并非穷尽式，也不代表任何优先顺序：

Using AI More Likely as a Tool

更倾向于将人工智能作为工具使用

The following application scenarios use AI more likely supportive to their general research process:

中文译文：以下应用场景更倾向于把人工智能作为一般研究流程中的辅助工具：

Original English table / 英文原表

Application Scenarios	*Implementation Examples*
Scientific Writing • Spell checking • Translation support • Managing citations • Formatting of text and/or bibliography • Text improvement • Text coherence improvement	• Grammarly • Writefull • DeepL • ChatAI of GWDG
Literature Review • Literature search • Text summary • Text explanation	• Elicit • ORKG Ask • Research Rabbit • Semantic Scholar
Scientific Brainstorming • Methodology brainstorming: Weighting pro and cons of a selected research method (validity, reliability, applicability) • Brainstorming limitations: Pointing to possible limitations and constraints of a study • Brainstorming biases: Highlighting potential biases in historical sources and interpretations	• ChatAI of GWDG (e.g., with Deepseek R1) • OpenAI Deep Research • Claude
Coding Assistance • Finding bugs: Identifying incorrect and/or incomplete code, and offering how to improve code • Code suggestions and solutions: Improving code, and/or providing code solutions • Data visualization: Providing code for data visualization	• ChatAI of GWDG (e.g., with Qwen 2.5 Coder 32B Instruct) • Copilot • GPT4All (has many coding models)
Data Exploration • Checking on data completeness • Pointing to errors or unusual patterns in data	• Pandas AI

中文表格译文：“人工智能作为工具”的应用场景

应用场景	具体任务	实施示例
科学写作	拼写检查；翻译支持；管理引文；文本和/或参考文献格式整理；文本改进；提升文本连贯性	Grammarly；Writefull；DeepL；GWDG ChatAI
文献综述	文献检索；文本摘要；文本解释	Elicit；ORKG Ask；Research Rabbit；Semantic Scholar
学术头脑风暴	方法论头脑风暴：权衡某一研究方法的优缺点（有效性、可靠性、适用性）；局限性头脑风暴：指出研究可能存在的局限和约束；偏差头脑风暴：提示历史史料与解释中的潜在偏差	GWDG ChatAI（如 DeepSeek R1）；OpenAI Deep Research；Claude
编程辅助	查找错误：识别错误或不完整代码并提出改	GWDG ChatAI（如 Qwen 2.5 Coder 32B

	进方式；代码建议与解决方案；数据可视化代码	Instruct）；Copilot；GPT4All
数据探索	检查数据完整性；指出数据中的错误或异常模式	Pandas AI

Using AI More Likely as a Method

更倾向于将人工智能作为方法使用

The following application scenarios use AI more likely as central part of their research process:

中文译文：以下应用场景更倾向于把人工智能作为研究流程的核心组成部分：

Original English table / 英文原表

Application Scenarios	*Implementation Examples*
Natural Language Processing (NLP) Tasks • Data Cleaning: OCR post-correction, preparing data for further qualitative or quantitative analysis • Text Extraction: Extracting relevant text from digitized sources to create suitable corpora tailored to the research question • Entities Extraction: Assisting in extracting and linking entities (e.g., individuals, organizations, places) and finding relations between them • Text Classification: Supporting text classification tasks such as sentiment analysis or genre classification. • Tagging: Tagging texts with relevant categories or labels.	• HuggingFace (e.g., NuExtract for information extraction) • GWDG models via API • Ollama (e.g., Llama family models)
Computer Vision • Image Extraction: Detecting and extracting images within digitized historical sources, fastening work flows via Multimodal Large Language Models (MLLMs) • OCR: Transcribing historical documents • Image Analysis: Analyzing image composition and content via MLLMs or Large Vision Models (LVMs)	• ChatAI of GWDG (e.g., OpenGVLab) • GWDG models via API • Google Coud AI • HuggingFace Model Caracal • Detectron 2
Method Development • Model training: Developing specialized models trained on highly specific datasets. • Comparative experiments: Systematic experimentation with different models on similar tasks.	• HuggingFace • Overiew in "Awesome List LLMs"

中文表格译文：“人工智能作为方法”的应用场景

应用场景	具体任务	实施示例
自然语言处理（NLP）任务	数据清理：OCR 后校正，为进一步的定性或定量分析准备数据；文本提取：从数字化史料中提取与研究问题相关的文本，建立合适语料库；实体提取：辅助识别并关联人物、机构、地点等实体并发现其关系；文本分类：支持情感分析、体裁分类等任务；标注：为文本添加相关类别或标签	HuggingFace（如 NuExtract）；通过 API 使用 GWDG 模型；Ollama（如 Llama 系列模型）
计算机视觉	图像提取：检测并提取数字化历史材料中的图像，借助多模态大型语言模型	GWDG ChatAI（如 OpenGVLab）；GWDG API；Google Cloud AI；

	（MLLMs）加快工作流程；OCR：转录历史文献；图像分析：使用 MLLMs 或大型视觉模型（LVMs）分析图像构图与内容	HuggingFace Caracal；Detectron 2
方法开发	模型训练：在高度特定的数据集上开发专门模型；比较实验：在相似任务上系统比较不同模型	HuggingFace；“Awesome List LLMs”概览

Individual and Institutional Responsibilities

个人与机构责任

Responsible AI Integration in historical research requires both thorough understanding of source materials and AI expertise. These requirements manifest at individual and institutional levels, creating a framework for effective implementation.

中文译文：在历史研究中负责任地整合人工智能，既需要对史料本身有深入理解，也需要相应的人工智能专业知识。这些要求同时体现在个人层面和机构层面，并共同构成有效实施人工智能的框架。

When Using AI as a Tool

当人工智能作为工具时

At the individual level, researchers using AI as a supporting tool need basic AI literacy to effectively utilize these technologies while maintaining critical awareness of their limitations. While user-friendly AI tools make routine tasks more accessible, researchers must still exercise careful judgment about their appropriate application and potential impacts on research outcomes.

中文译文：在个人层面，将人工智能作为辅助工具的研究者需要具备基本的人工智能素养，既能够有效使用这些技术，也要对其局限保持批判性认识。即使用户友好的人工智能工具降低了日常任务的使用门槛，研究者仍必须谨慎判断这些工具是否适用于具体任务，以及它们可能对研究结果造成何种影响。

At the institutional level, important requirements are clear guidelines for tool selection, data security protocols, and ethical considerations specific to AI-assisted research tools. This includes support for commonly used AI applications while maintaining research integrity.

中文译文：在机构层面，关键要求包括：为工具选择制定清晰准则，建立数据安全规范，并针对人工智能辅助研究工具提出伦理要求。机构既应为常用人工智能应用提供支持，也必须维护研究诚信。

When Using AI as a Method

当人工智能作为方法时

At the individual level, researchers are responsible to develop deeper technical expertise and/or to establish substantial collaborations with AI specialists. This requires understanding not just how to use AI, but how it fundamentally affects research design, data analysis, and interpretation of results.

中文译文：在个人层面，研究者有责任发展更深入的技术能力，和/或与人工智能专家建立实质性合作。这不仅要求研究者知道如何使用人工智能，还要理解人工智能会如何从根本上影响研究设计、数据分析以及研究结果的解释。

At the institutional level, key requirements are clear guidelines that prioritizes data security and ethical AI usage as well as recommendations on how to get the needed technical infrastructure.

中文译文：在机构层面，关键要求是制定明确的规范，把数据安全和人工智能的伦理使用置于优先位置，同时提供如何获得必要技术基础设施的建议。

Common Ground at the DH Lab at IEG Mainz

IEG Mainz 数字人文实验室的共同基础

At IEG Mainz, collaboration between domain experts and technical specialists has led to a robust environment for methodological innovation. Both approaches, AI as a tool or a method, benefit from the DH Lab's collaborative framework within the IEG, which brings together domain expertise and technical knowledge. As methodological best practices for implementing LLMs in historical research continue to emerge, this integrated approach provides a strong foundation for developing robust research standards while maintaining scholarly rigor.

中文译文：在 IEG Mainz，领域专家与技术专家之间的合作形成了有利于方法创新的稳健环境。无论把人工智能作为工具还是作为方法，都能从 IEG 内部 DH Lab 的协作框架中受益，因为这一框架把领域知识与技术知识

结合起来。随着大型语言模型在历史研究中的方法论最佳实践不断发展，这种整合式路径能够在保持学术严谨性的同时，为建立稳健的研究标准提供坚实基础。

Through meaningful exchange and clear guidelines, researchers can effectively navigate both technical and methodological complexities, whether using AI as a supplementary tool or as a core research method.

中文译文：通过有实质内容的交流与清晰规范，无论研究者把人工智能作为辅助工具，还是作为核心研究方法，都能够更有效地处理技术与方法论层面的复杂问题。

What You Need to Know Before to Start?
开始之前需要了解什么?

Overall Questions to Ask
需要先问自己的总体问题

Before using AI technology as a tool (or method), we recommend to ask yourself key questions that should help create awareness of both capabilities and limitations while developing the practical skills needed to integrate AI technology into your own research agenda or research methodology:

中文译文：在把人工智能技术作为工具（或方法）使用之前，作者建议研究者先问自己一些关键问题。这些问题既有助于认识人工智能的能力与局限，也有助于形成将人工智能整合进个人研究议程或研究方法所需的实践技能：

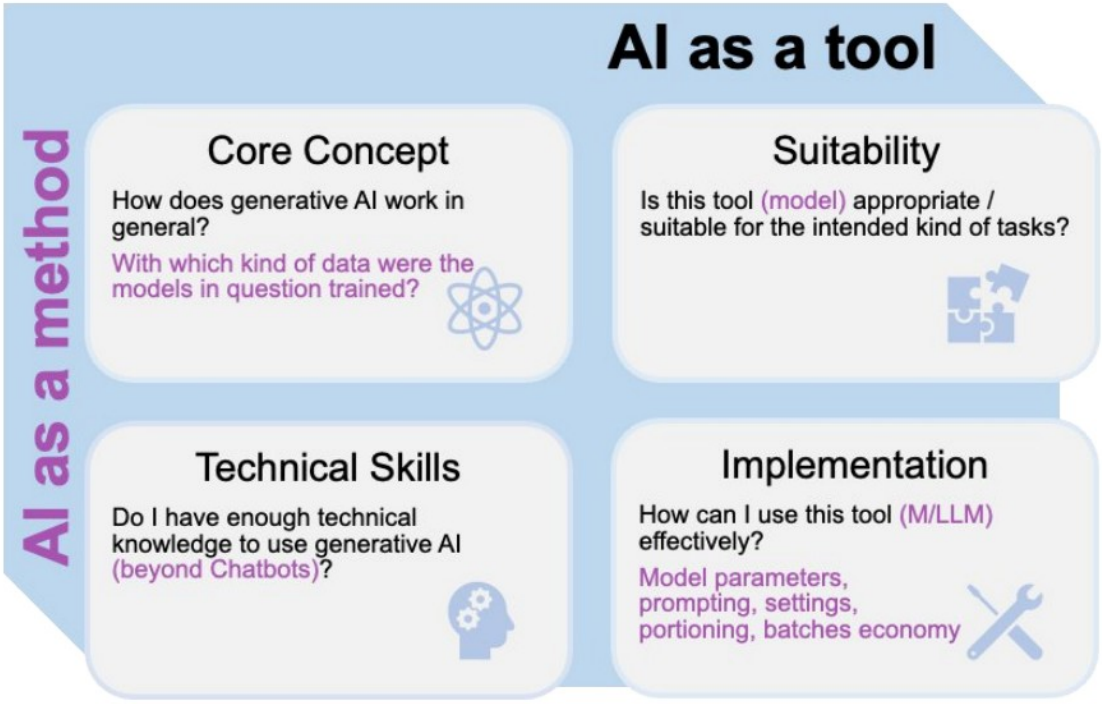


Fig. 1: Questions to ask before considering the use of generative AI for research (Image: S. Oberbichler & C. Petz 2025). The violet text refers to questions relevant when using AI as a method.

Fig. 1: Questions to ask before considering the use of generative AI for research (Image: S. Oberbichler & C. Petz 2025). The violet text refers to questions relevant when using AI as a method.

中文译文：图 1：在考虑将生成式人工智能用于研究之前需要提出的问题（图：S. Oberbichler & C. Petz，2025）。紫色文字表示与“人工智能作为方法”相关的问题。

Risk Assessment
风险评估

To ensure optimal results, we need to balance performance, stability, legal and ethical concerns and ecological costs. This means, too, if other less cost-intensive methods than LLMs would provide similar performances, these should be favored following the idea of minimal computing. However, low technical hurdles and project time restraints do not replace the necessary reflection on the technology itself as part of good research practice.

中文译文：为了获得最佳结果，需要在性能、稳定性、法律与伦理问题以及生态成本之间取得平衡。这也意味着，如果某些成本低于大型语言模型的方法能够提供相近的表现，那么依据“最小计算”的理念，应优先采用这些方法。与此同时，技术门槛较低或项目时间有限，都不能替代对技术本身的必要反思；这种反思是良好研究实践的一部分。

While EU AI Act outlines regulations for safe AI usage, it has restricted applicability for the historical studies, but still provides a great starting point for reflection. In order to evaluate whether to use a LLM-based tool (or method), please assess the following risks:

中文译文：虽然《欧盟人工智能法案》（EU AI Act）规定了安全使用人工智能的相关规则，但这些规定在历史研究中的直接适用范围有限。不过，它仍然提供了一个很好的反思起点。为了判断是否应使用基于大型语言模型的工具（或方法），需要评估以下风险：

1 See Infobox on p. 10.

中文译文：注 1：见原文第 10 页的信息框。

Societal Risks and Limits to the Safe, Recommended, or Legal Applicability of AI
社会风险，以及人工智能安全、适宜或合法应用的边界

Suitability: Was the tool (or model) designed for the intended application?

中文译文：适用性：该工具（或模型）最初是否就是为你计划开展的应用而设计的?

Legal Dimensions: Are GDPR Guidelines followed? Does usage conform to the EU AI Act? Are the sources used open-access, closed-access, licensed, and/or contain sensitive data?

中文译文：法律维度：是否遵守 GDPR 指南？使用方式是否符合《欧盟人工智能法案》？所使用的史料是开放获取、封闭获取、受许可限制，还是包含敏感数据?

This influences your decision on which tools (or models) you should use! Note, that data security and privacy concerns arise when processing prompts on third-party-owned services. There, inputs, prompts, and requests are sent to third-party-owned servers, which might not adhere to European privacy regulations.

中文译文：这些因素会直接影响你应选择何种工具（或模型）。需要注意，当提示词通过第三方所有的服务进行处理时，会产生数据安全与隐私问题；输入内容、提示词和请求会被发送到第三方服务器，而这些服务器未必遵守欧洲隐私法规。

Applicability: What are the limits to the safe, recommended, or legal applicability of AI? In the context of the historical studies, it is essential to know the limitations of using AI tools when, e.g.:

中文译文：应用边界：人工智能在安全、适宜或合法使用方面有哪些限制？在历史研究语境中，尤其需要了解在下列情形中使用人工智能工具的限制：

- grading student papers
  - 给学生论文评分
- writing peer reviews
  - 撰写同行评审意见
- writing grant proposals
  - 撰写科研资助申请
- reviewing grant proposals
  - 评审科研资助申请

Additional Reading: Ghent University (2024): GDPR: What should I take into account when developing or using AI? AND Forschungsteam ZGKI (2025): EU AI ACT Kompass. Executive Committee of the DFG (2023): Statement on the Influence of Generative Models of Text and Image Creation on Science and the Humanities and on the DFG's Funding Activities.

中文译文：延伸阅读：Ghent University（2024）《开发或使用人工智能时应如何考虑 GDPR？》；Forschungsteam ZGKI（2025）《EU AI ACT Kompass》；DFG 执行委员会（2023）《关于生成式文本与图像模型对科学、人文学科及 DFG 资助活动影响的声明》。

Ecological and Economical Consequences
生态与经济后果

Immediate costs:

中文译文：直接成本：

Original English table / 英文原表

Immediate costs:

AI as a tool	*AI as a method*
Costs of using the tool (e.g., ChatGPT membership vs. free Shibboleth access to ChatAI)	Costs of using or buying model access (API), computing time access (GPU/CPU), prompt or token access, ...

Hidden costs: Which damages to the environment are related to the use of the tool (or model)?

中文表格译文：直接成本与隐性成本

	人工智能作为工具	人工智能作为方法
直接成本	使用工具的费用（例如 ChatGPT 会员费，与通过 Shibboleth 免费访问 ChatAI 相比）	使用或购买模型访问权限（API）的费用、计算时间（GPU/CPU）、提示词或 token 访问费用等
隐性成本	该工具对环境造成了哪些相关损害?	该模型对环境造成了哪些相关损害?

Additional Reading: Li et al (2023): Making AI Less "Thirsty": Uncovering and Addressing the Secret Water Footprint of AI Models AND Carbon Footprint Calculator for Machine Learning

中文译文：延伸阅读：Li 等（2023）《让人工智能不再那么“口渴”：揭示并应对人工智能模型隐秘的水足迹》；机器学习碳足迹计算器。

Centrality of LLMs in the proposed project
大型语言模型在拟议项目中的核心程度

Necessity: Is it necessary to use a LLM-based tool (or method) for this project? Will you do systematic experiments with LLMs in this project?

中文译文：必要性：该项目是否真的有必要使用基于大型语言模型的工具（或方法）？项目是否会对大型语言模型开展系统性实验?

Inevitability: Would alternative tools (or methods) provide similar results and with similar costs? If so, these alternative methods might be preferred to using LLM-based tools or methods. This follows the idea of minimal computing.

中文译文：不可替代性：其他工具（或方法）是否能够以相近成本获得相似结果？如果可以，那么依据“最小计算”的理念，可能应优先选择这些替代方法，而不是基于大型语言模型的工具或方法。

Key Concept: Minimal Computing in AI Research
核心概念：人工智能研究中的“最小计算”

The concept of minimal computing refers to environmentally conscious usage of AI by:

中文译文：“最小计算”指以环境意识为基础使用人工智能，具体包括：

- Using LLMs selectively when they demonstrate clear performance advantages

· 只有在大型语言模型显示出明确性能优势时才有选择地使用它们；

• Conducting systematic experiments to validate necessity.

· 通过系统实验验证其使用是否确有必要；

• Employing specialized, task-specific models rather than general-purpose ones.

· 优先使用面向特定任务的专用模型，而不是通用模型；

• Using models as small as possible.

· 尽可能使用更小的模型。

This approach optimizes computational resources while maintaining research effectiveness.

中文译文：这种路径在保持研究有效性的同时，可以优化计算资源的使用。

Recommended Reading: Digital Humanities Climate Coalition (2025): Minimal Computing

中文译文：推荐阅读：Digital Humanities Climate Coalition（2025）《Minimal Computing》。

Taking Over Responsibility When Using AI as a Tool and as a Method

将人工智能作为工具或方法使用时承担责任

Using LLM-based tools or methods require the need to take over responsibility and human operational oversight by active engagement ("human-in-the-loop") in the following steps:

中文译文：使用基于大型语言模型的工具或方法，需要研究者承担责任，并通过主动参与保持人工操作监督，即在以下环节中贯彻"人在回路"（human-in-the-loop）：

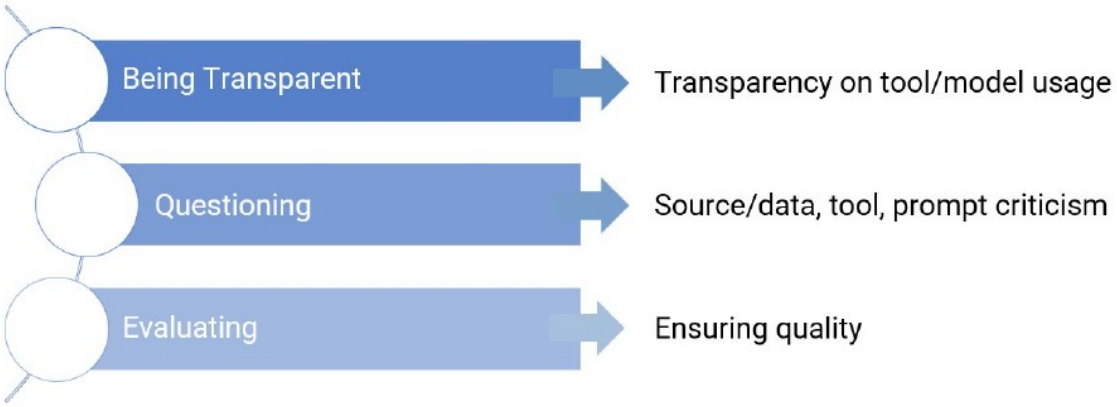


Fig. 2: Reflection on steps in the AI-assisted research process (Image: S. Oberbichler & C. Petz 2025).

中文译文：图 2：对人工智能辅助研究流程各步骤的反思（图：S. Oberbichler & C. Petz，2025）。

This ensures methodological rigor while addressing AI-specific research challenges.

中文译文：这样做既能保证方法论严谨性，也能应对人工智能研究所特有的挑战。

Being Transparent

保持透明

When incorporating AI into academic research, whether as a tool (or method), adherence to rigorous academic standards is essential. As with any research instrument or methodology, transparency about AI usage is fundamental to maintaining scholarly integrity. These measures increase the reproducibility of findings. But as LLMs are not deterministic, there are limits to this!

中文译文：无论把人工智能作为工具（或方法）纳入学术研究，都必须遵守严格的学术标准。和任何研究工具或方法一样，对人工智能的使用保持透明，是维护学术诚信的基本要求。这些做法能够提高研究发现的可复现性；不过，由于大型语言模型并非确定性系统，可复现性本身也存在一定限度。

Original English table / 英文原表

rigorous academic standards is essential. As with any research instrument or methodology, transparency about AI usage is fundamental to maintaining scholarly integrity. These measures increase the reproducibility of findings. But as LLMs are not deterministic, there are limits to this!

AI as a tool	AI as a method
Crediting which tool (and possibly which model version) was used for which kind of tasks, including which prompts if applicable.	Crediting which model of which origin was used for which kind of tasks • Reporting technical specifications: Model version, GPU, .. • Documenting model parameters

中文表格译文：透明性要求

人工智能作为工具	人工智能作为方法
说明在何种任务中使用了哪一种工具，并在可能的情况下说明模型版本；如适用，还应记录使用了哪些提示词。	说明在何种任务中使用了何种来源的模型；报告技术规格（如模型版本、GPU 等）；记录模型参数、设置、分批大小、提示词等。

Additional Resources: Leibniz Association (2024): Empfehlung zur Sicherung der guten wissenschaftlichen Praxis beim Umgang mit Künstlicher Intelligenz AND Oberbichler (2024): AI Model Research Documentation Sheet (AIRDocS)

中文译文：补充资源：Leibniz Association（2024）《关于在使用人工智能时保障良好科学实践的建议》；Oberbichler（2024）《人工智能模型研究文档表（AIRDocS）》。

Questioning Inputs

审视输入

When working with AI models, source criticism, data criticism, and tool (or method) criticism are very much interconnected. This means that in order to critically reflect, for example, the appropriateness or applicability of a model, it is essential to examine your data as well as the data the model was trained on for a specific task. First, we take a look at the left half of the epistemological framework as shown in Fig. 3:

中文译文：在使用人工智能模型时，史料批判、数据批判与工具（或方法）批判彼此紧密相连。这意味着，为了批判性反思一个模型是否适当、是否适用于具体任务，不仅要检查自己的数据，也需要考察模型针对该任务所使用的训练数据。首先，作者从图 3 所示认识论框架的左半部分开始讨论：

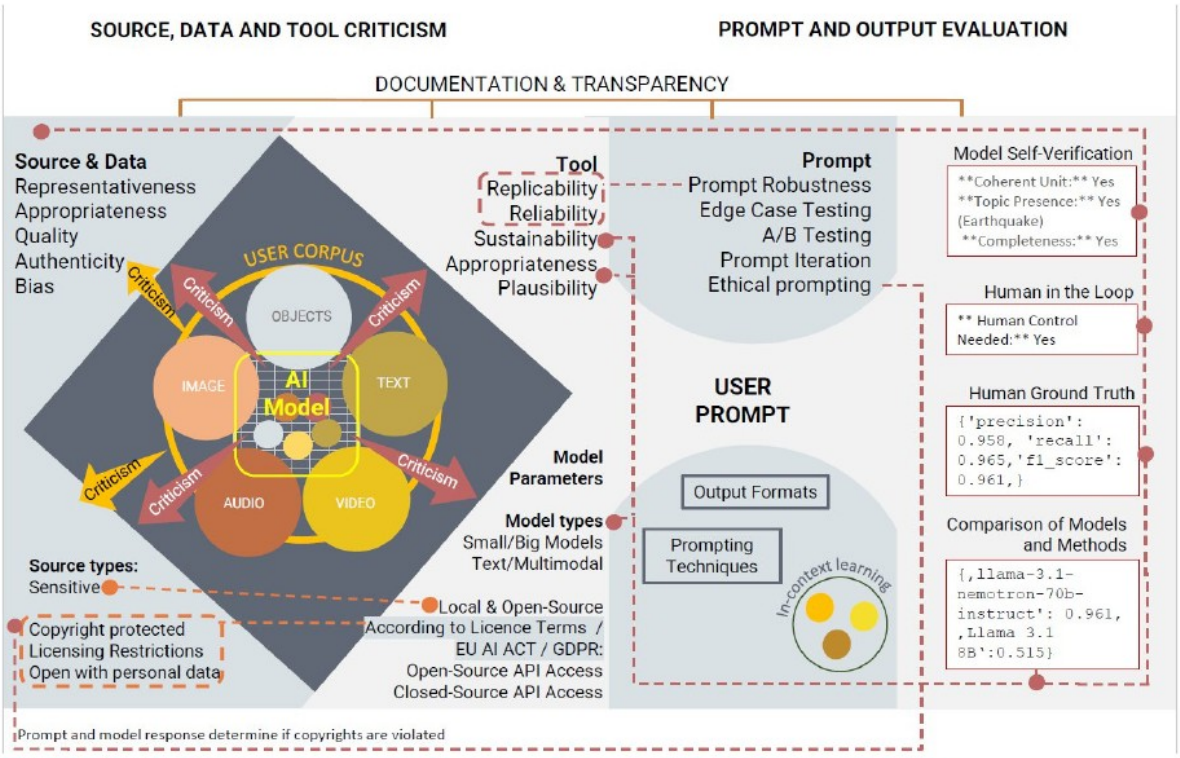


Fig. 3 Epistemological framework for AI-assisted research. (Image: S. Oberbichler & C. Petz 2024).

Fig. 3: Epistemological framework for AI-assisted research (Image: S. Oberbichler & C. Petz 2024).

中文译文：图 3：人工智能辅助研究的认识论框架（图：S. Oberbichler & C. Petz，2024）。

Extending the established critical reflection of source criticism to data requires examining four key areas:

中文译文：将传统史料批判中的批判性反思扩展到数据层面，需要考察四个关键领域：

Critical reflection on source & data examines provenance, content, context, and formal aspects like data types. This includes understanding how the dataset was collected, what transformations were applied, and whether the collection methods introduced systematic errors.

中文译文：对史料与数据的批判性反思需要考察其来源、内容、语境以及数据类型等形式特征。其中包括理解数据集如何被收集、经历了哪些转换处理，以及收集方式是否引入了系统性错误。

Critical reflection on data biases examines inherent perspectives, mis-/representation issues, and possible discrimination. This involves examining how different groups, languages, and domains are represented in the data, identifying potentially harmful categorizations, and understanding whose perspectives are privileged or excluded in the dataset.

中文译文：对数据偏差的批判性反思，需要考察数据中固有的立场、错误呈现或代表性不足问题，以及可能存在的歧视。这包括分析不同群体、语言和领域在数据中如何被呈现，识别可能造成伤害的分类方式，并理解哪些人的视角在数据集中被优先呈现，哪些视角被排除。

Critical reflection on tools and models investigates the technical implementation as well as their appropriateness. This means examining the model architecture, understanding how the tool processes and transforms data, and considering the consequences of its deployment in different contexts (applicability).

中文译文：对工具与模型的批判性反思，既要考察其技术实现，也要考察其适用性。这意味着需要审视模型架构，理解工具如何处理和转换数据，并考虑它在不同情境中部署后可能产生的后果，也就是其具体适用性。

Original English table / 英文原表

AI as a tool	AI as a method
Tool criticism • Appropriateness of the tool for the intended tasks and data types: What was this tool trained on and what was it trained to do?	Model criticism: • Appropriateness of the model for the intended tasks and data types: What was this model trained on and what was it trained to do? • Using specialized and as small as possible models following the principle of minimal computing in AI-based research • Assessing the influence of the training data on the results and representation

中文表格译文：工具批判与模型批判

人工智能作为工具	人工智能作为方法
工具批判：判断工具是否适用于预期任务和数据类型。需要追问：这个工具是基于什么训练的？它原本被训练来做什么？	模型批判：判断模型是否适用于预期任务和数据类型。需要追问：模型是基于什么训练的？它原本被训练来做什么？遵循人工智能研究中的最小计算原则，优先使用专用且尽可能小的模型；评估训练数据对结果与代表性的影响。

Adhering to legal considerations of data usage requires careful attention to data privacy regulations, copyright restrictions, and usage limitations when working with sensitive or private or confidential material, copyrighted content, licensed data, restricted information, including unpublished works or student submissions.

中文译文：在数据使用方面遵守法律要求，意味着在处理敏感、私人或保密材料、受版权保护的内容、受许可限制的数据、受限制信息，以及未发表作品或学生提交材料时，必须认真关注数据隐私法规、版权限制以及使用范围限制。

- Awareness about legal liabilities through possible violations of privacy, licenses, confidentiality, and other legal limitations of datasets used. Content uploaded (including input, prompts, requests) to Chatbots and other cloud-based AI systems can be potentially exposed to unauthorized access, commercially reused, sold, and linked with other personal or sensitive information.
  - 需要意识到因违反隐私、许可协议、保密义务以及所用数据集的其他法律限制而产生的法律责任。上传到聊天机器人和其他云端人工智能系统的内容（包括输入、提示词和请求）可能面临未经授权访问、商业再利用、出售，或与其他个人信息及敏感信息关联的风险。
- Awareness about copyright violations through harmful prompts or predatory models: Note: Meta-analysis & summaries of copyrighted material are usually legal, but reproducing copyrighted material is a violation of the copyright. Note: In Germany, the “Wissenschaftsschranke” allows researchers to use copyrighted works for scientific purposes without needing explicit permission. This is vital for fostering academic research, as it enables access to a broader range of materials for analysis and study. However, researchers must be aware of potential conflicts between the “Wissenschaftsschranke” and licensing agreements. Generally, regulations by archives should be respected in order to allow good collaboration.

· 需要意识到有害提示词或掠夺性模型可能造成版权侵权。原文特别指出：对受版权保护材料进行元分析和摘要通常是合法的，但直接复制受版权保护材料会构成版权侵权。在德国，“Wissenschaftsschranke”（科学研究例外条款）允许研究者出于科学目的使用受版权保护作品而无需获得明确许可，这对学术研究非常重要，因为它扩大了可供分析和研究的材料范围。不过，研究者需要注意该条款与具体许可协议之间可能发生的冲突。总体而言，应尊重档案机构制定的规定，以维持良好的合作关系。

2 See also Chapter: “What do I need to know before to start?” (p. 6-7)

中文译文：注 2：另见原文“开始之前需要了解什么？”一节（第 6-7 页）。

Original English table / 英文原表

Security and privacy concerns: Awareness about legal liabilities of sensitive information and copyrighted materials when using cloud-based AI tools	Security and privacy concerns: Sensitive and/or copyright protected material should be processed via local models, or European-based cloud servers. Commercially used platforms should be avoided.

中文表格译文：安全与隐私问题

人工智能作为工具	人工智能作为方法
使用云端人工智能工具时，应意识到敏感信息与受版权保护材料所涉及的法律责任。	敏感信息和/或受版权保护材料应通过本地模型或位于欧洲的云服务器处理；应避免使用商业化平台。

Additional Resources: Markert (v.0.1): Entscheidungsbaum zu urheberrechtlichen Aspekten der Nutzung von Digitalisaten; Anne Burkhardt (2025): KI und Globale (Un)Gerechtigkeit: Ethische Perspektiven auf Künstliche Intelligenz; Navigli et al. (2023): Biases in Large Language Models: Origins, Inventory, and Discussion; Ferrer et al. (2021): Bias and Discrimination in AI: A cross-disciplinary perspective; Mauermann & Oberbichler (2025): LLM Biases: Expected and Unexpected Model Design Effects in Historical Newspaper Article Extraction on the Messina Earthquake.

中文译文：补充资源：Markert（v0.1）关于数字化材料使用中版权问题的决策树；Anne Burkhardt（2025）关于人工智能与全球（不）公正的伦理讨论；Navigli 等（2023）关于大型语言模型偏差的来源、分类与讨论；Ferrer 等（2021）关于人工智能中的偏差与歧视；Mauermann 与 Oberbichler（2025）关于大型语言模型在墨西拿地震历史报纸文章提取中的预期与非预期模型设计效应。

Evaluating Outputs

评估输出

The right half of the epistemological framework (Fig. 3) concerns the output evaluation. It is recommended to critically reflect on AI tools' (or models') roles and limitations while maintaining responsibility for output verification. This includes documenting how AI systems influence our research processes, understanding tool (or model) constraints, and validating or verifying AI-generated outputs (compare to right half of Fig. 2).

中文译文：认识论框架（图 3）的右半部分讨论输出评估。作者建议，在持续承担输出核验责任的同时，应批判性反思人工智能工具（或模型）的作用与局限。具体包括：记录人工智能系统如何影响研究流程，理解工具（或模型）的约束条件，并对人工智能生成的输出进行验证或核查（可与图 2 右侧相对照）。

- E.g., comparing generated summaries of text
  - 例如，比较人工智能生成的文本摘要；
- E.g., if generated code actually works as intended
  - 例如，检查生成的代码是否真的按照预期运行；
- E.g., if the code works consistently well
  - 例如，检查代码是否能够持续稳定地运行。

- E.g., if generated code actually works as intended
- E.g., if the code works consistently well

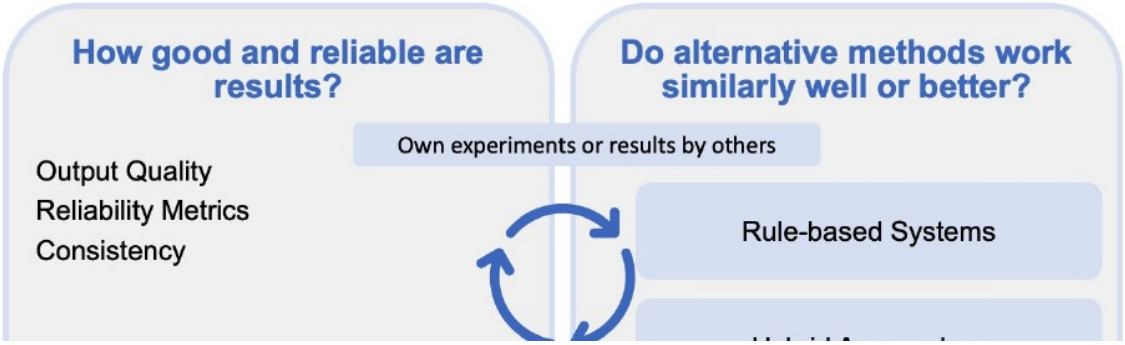


Fig. 4: Performance Feedback Loop (Image: S. Oberbichler & C. Petz 2024).

中文译文：图 4：性能反馈循环（图：S. Oberbichler & C. Petz，2024）。

Considering limitations of the reliability of the output includes being aware about:

中文译文：在考虑输出可靠性的局限时，需要意识到：

- replication of biases within the (tool or model training, and your own source) dataset into the results. The reflection on biases is a necessary extension of the source criticism of your own dataset.
  - 工具或模型训练数据以及研究者自身史料数据中的偏差，可能被复制到最终结果中。因此，对偏差的反思，是对自身数据集开展史料批判时不可缺少的延伸。
- confabulations ('hallucinations'), when the tool (or model) is inventing 'facts'. LLMs are "stochastic parrots," this means they tend to recreate their training input with a certain probability.
  - 模型可能产生"虚构"（即"幻觉"），也就是工具或模型自行编造所谓"事实"。大型语言模型常被称为"随机鹦鹉"（stochastic parrots），意指它们会以一定概率重新组合和再现训练输入中的内容。

Original English table / 英文原表

AI as a tool	AI as a method
Comparison of different tools	Systematic experimentation: • Optimization of model parameters, settings, temperature, batches, prompts, ... • Prompt iteration & prompt engineering (“prompt criticism”) • Note: Strategies for prompt engineering depend on the model used • Portioning (“batches economy”)
Quality of results: • Qualitative assessment of the output • Checking each step of the tool (if applicable)	Quality of results: • Evaluation and robustness tests of results + documentation • Explainability (“Explainable AI”) • Reliability & plausibility • In general: How good is good enough? Do you need to stop at 90% accuracy? • Balancing quality vs time resources • Performance Feedback Loop

中文表格译文：输出评估：工具与方法

人工智能作为工具	人工智能作为方法
比较不同工具。	开展系统实验：优化模型参数、设置、temperature、批次、提示词等；迭代提示词并进行提示工程（“提示批判”）；注意提示工程策略取决于所用模型；进行分批处理（“批次经济”）。
结果质量：对输出进行定性评估；如适用，检查工具运行的每一个步骤。	结果质量：开展结果评估与稳健性测试并记录；考察可解释性（Explainable AI）；检查可靠性与合理性；追问“多好才算足够好”，是否需要以90%准确率为停止点；平衡质量与时间资源；使用性能反馈循环；系统测试后检查替代方法是否能获得相近或更好效果，如能，则依据最小计算理念优先采用替代方法。

Recommendations for Implementation

实施建议

Open-Weight, Open-Source and Closed-Source Models

开放权重、开源与闭源模型

LLMs exist in difference sizes and accessibility and/or transparency levels. The explanations below give insight into the difference between open-weight, open-source, and closed-source models:

中文译文：大型语言模型具有不同规模，同时在可访问性和/或透明度方面也存在差异。下面的说明用于区分开放权重、开源和闭源模型：

Original English table / 英文原表

models:

	What is Available	*What Can be Done*
Open-Source Models	Everything: code, training method, data	These models allow for full modification and insight into how it works. These models allow to have a insight into the data the model is trained on. They can be used locally.
Open-Weight Models	Just the trained model parameters	These models can be fine-tuned and adapted within limits. They can be used locally.
Closed-Source Models	Only the interface/API	These models are proprietary, controlled and maintained within a closed environment such as Chatbots. Access to the underlying code and training data is not possible, often making these models black boxes to external users. These models cannot be used locally.

中文表格译文：开放权重、开源与闭源模型的差别

模型类型	可获得内容	可以做什么
开源模型	全部内容：代码、训练方法、数据	可以完整修改并深入了解其工作机制；可以了解模型所使用的训练数据；可以在本地运行。
开放权重模型	仅提供训练后的模型参数	可以在一定范围内进行微调和适配；可以在本地运行。
闭源模型	仅提供界面/API	模型属于专有系统，在封闭环境（如聊天机器人）中被控制和维护；无法访问底层代码

		和训练数据，因此对外部用户而言往往是“黑箱”；无法在本地运行。

We recommend using fully open-source models where possible, as they allow critical examination of training data and complete transparency of the model architecture. Models like Tülu 3 from The Allen Institute for AI or from LLM360 enable to better understand potential biases and limitations. Running these models locally also ensures that your research data, prompts, and outputs remain private and under your control.

中文译文：作者建议在条件允许时优先使用完全开源的模型，因为开源模型允许研究者批判性检查训练数据，并获得对模型架构的完整透明度。例如 Allen Institute for AI 的 Tülu 3 或 LLM360 提供的模型，有助于研究者更好理解潜在偏差与局限。在本地运行这些模型，也可以确保研究数据、提示词和输出保持私密并由研究者自己掌控。

When open-source models aren't feasible, open-weight models can provide a good alternative. Open-weight models in various sizes, such as the Llama models by Meta, can be run locally to ensure full privacy.

中文译文：当开源模型不可行时，开放权重模型可以成为较好的替代方案。不同规模的开放权重模型，例如 Meta 的 Llama 系列，可以在本地运行，从而保障完整的数据隐私。

For cases where closed-source models such as Google Cloud AI are the best option, we recommend using APIs rather than web interfaces, as this provides more control over the interaction and better integration possibilities for research workflows. Note, that data security and privacy concerns arise when processing prompts on third-party-owned services. Input, prompts and requests are sent to the service, which might not adhere to European privacy regulations.

中文译文：如果 Google Cloud AI 等闭源模型确实是最佳选择，作者建议优先通过 API 而非网页界面使用，因为 API 能够为交互提供更多控制，也更便于整合进研究工作流。需要注意的是，在第三方所有的服务上处理提示词时会产生数据安全与隐私问题；输入、提示词和请求会被发送给该服务，而服务提供方未必遵守欧洲隐私法规。

3 Compare to chapter on Legal Dimension of Risk Analysis, p. 6-7.

中文译文：注 3：参见原文第 6-7 页风险分析中“法律维度”部分。

Accessing Large Language Models Beyond Commercial Tools
在商业工具之外访问大型语言模型

Large Language Models can be accessed beyond integrated components of specific research tools (like literature search engines, or translation software such as DeepL) or commercial Chatbots like ChatGPT in three main ways:

中文译文：除了通过特定研究工具的内置组件（如文献检索引擎、DeepL 等翻译软件）或 ChatGPT 之类商业聊天机器人之外，研究者还可以主要通过以下三种方式访问大型语言模型：

- Locally via research Chat-Interfaces or research API’s for programmatic access and workflow integration.
  - 通过研究型聊天界面或研究 API 在本地或受控环境中访问，以便进行程序化调用和工作流整合；
- Through local installation on your own hardware (via chat-interfaces or programming environments).
  - 在自己的硬件上本地安装，通过聊天界面或编程环境运行；
- Through HPC clusters providing model and GPU access.
  - 通过提供模型与 GPU 访问权限的高性能计算（HPC）集群运行。

Below we list examples for each access point. While this list does not claim to be exhaustive, it should give a good overview of the available access options for researchers at the time this paper was published.

中文译文：下面列出每一种访问方式的示例。该列表并非穷尽，但能够较好概括本文发表时研究者可获得的主要访问选项。

Cloud-based Research Chat-Interfaces or Research API-Access

云端研究型聊天界面或研究 API 访问

The GWDG in Göttingen, for example, provides secure access to open-source and open-weight models in Germany. All services are hosted in academic data centers according to German data protection and data security directives.

中文译文：例如，位于哥廷根的 GWDG 在德国提供对开源模型和开放权重模型的安全访问。所有服务均托管于学术数据中心，并遵循德国的数据保护与数据安全规范。

- Their Chatbot ChatAI offers an easy and secure access to powerful generative AI. The intuitive interface allows users to chat directly with a selection of different AI models. Using your Shibboleth login credentials, the interface can be accessed free of charge.
  - 其聊天机器人 ChatAI 提供了便捷且安全的强大生成式人工智能访问方式。直观的界面允许用户直接与多种人工智能模型对话；使用 Shibboleth 登录凭据即可免费访问。
- API access for the usage of the models in programming environments can be requested (up to 200 calls per day).
  - 研究者还可以申请 API 访问权限，在编程环境中使用这些模型（每天最多 200 次调用）。

Locally Installed Chat-Interfaces

本地安装的聊天界面

Locally installed Desktop Apps allow the usage of large language models to run privately without local data ever leaving the private device. An example is GPT4ALL by Nomic, an open-source ecosystem used for integrating LLMs into consumer hardware without paying for a platform or hardware subscription.

中文译文：本地安装的桌面应用允许研究者以私密方式运行大型语言模型，使本地数据无需离开私人设备。Nomic 的 GPT4All 就是一个例子，它是开源生态系统，可将大型语言模型整合进普通消费级硬件，而不需要支付平台费用或硬件订阅费。

Model Running Platforms and Programming Environments

模型运行平台与编程环境

Ollama offers a powerful and user-friendly way to run various LLMs locally, supporting multiple models through both chat interfaces and programming environments. Ollama is an open-source project that serves as a powerful and user-friendly platform for running LLMs on your local machine.

中文译文：Ollama 提供了一种功能强大且用户友好的本地大型语言模型运行方式，可以通过聊天界面和编程环境支持多种模型。Ollama 是开源项目，可作为在本地计算机上运行大型语言模型的便捷平台。

HuggingFace provides comprehensive tools for model deployment:

中文译文：HuggingFace 提供了较为完整的模型部署工具：

- The Transformers library for implementing models in Python
  - 用于在 Python 中实现模型的 Transformers 库；
- Model Hub for accessing and downloading thousands of models
  - 可访问并下载数千种模型的 Model Hub；
- Specialized tools for local model optimization and deployment
  - 用于本地模型优化与部署的专用工具。

Anacondas provides on-premises deployment options for LLMs, enabling researchers to securely host and utilize LLMs locally.

中文译文：Anaconda 提供大型语言模型的本地部署选项，使研究者能够安全地在本地托管和使用大型语言模型。

Note: Limited computing power may require external CPU/GPU servers (e.g., Cloud-based GPU, renting GPU server, or buying a GPU server). For example, Google Colab Notebooks offer 20 hours of free GPU per week. GroqCloud offers free API access to a specific limit.

中文译文：注意：如果本地计算能力有限，可能需要使用外部 CPU/GPU 服务器，例如云端 GPU、租用 GPU 服务器或购买 GPU 服务器。原文举例指出，Google Colab Notebooks 每周提供 20 小时免费 GPU 使用时间；GroqCloud 则在一定额度内提供免费 API 访问。

HPC Clusters Providing Model Access

提供模型访问的 HPC 集群

When using mid- to large-size models that require more computing power, access to High Performance Computing (HPC) clusters can be an option, where you can externalize CPU/GPU usage. Usually, access is granted only after a formalized request:

中文译文：当使用需要更多计算资源的中型或大型模型时，可以考虑访问高性能计算（HPC）集群，把 CPU/GPU 运算转移到外部计算环境。通常需要通过正式申请才能获得访问权限：

- In Rhineland-Palatinate, the web-based interface MOGON OnDemand simplifies access to the HPC cluster by Allianz für Hochleistungsrechnen Rheinland-Pfalz (AHRP). They allocate a certain percentage of the central computing capacity available at the universities of Mainz (JGU) and Kaiserslautern-Landau (RPTU) in a shared pool for scientists and researchers in Rhineland-Palatinate.

  · 在莱茵兰-普法尔茨州，网页界面 MOGON OnDemand 简化了对 Rheinland-Pfalz 高性能计算联盟（AHRP）HPC 集群的访问。该联盟把美因茨大学（JGU）与凯泽斯劳滕-兰道大学（RPTU）中央计算能力中的一定比例放入共享资源池，供该州科学家和研究者使用。

- The GWDG (Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen) provides HPC resources and services for researchers at local, regional, and national levels.

  • GWDG（哥廷根科学数据处理有限公司）则为地方、区域和国家层面的研究者提供 HPC 资源与服务。

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